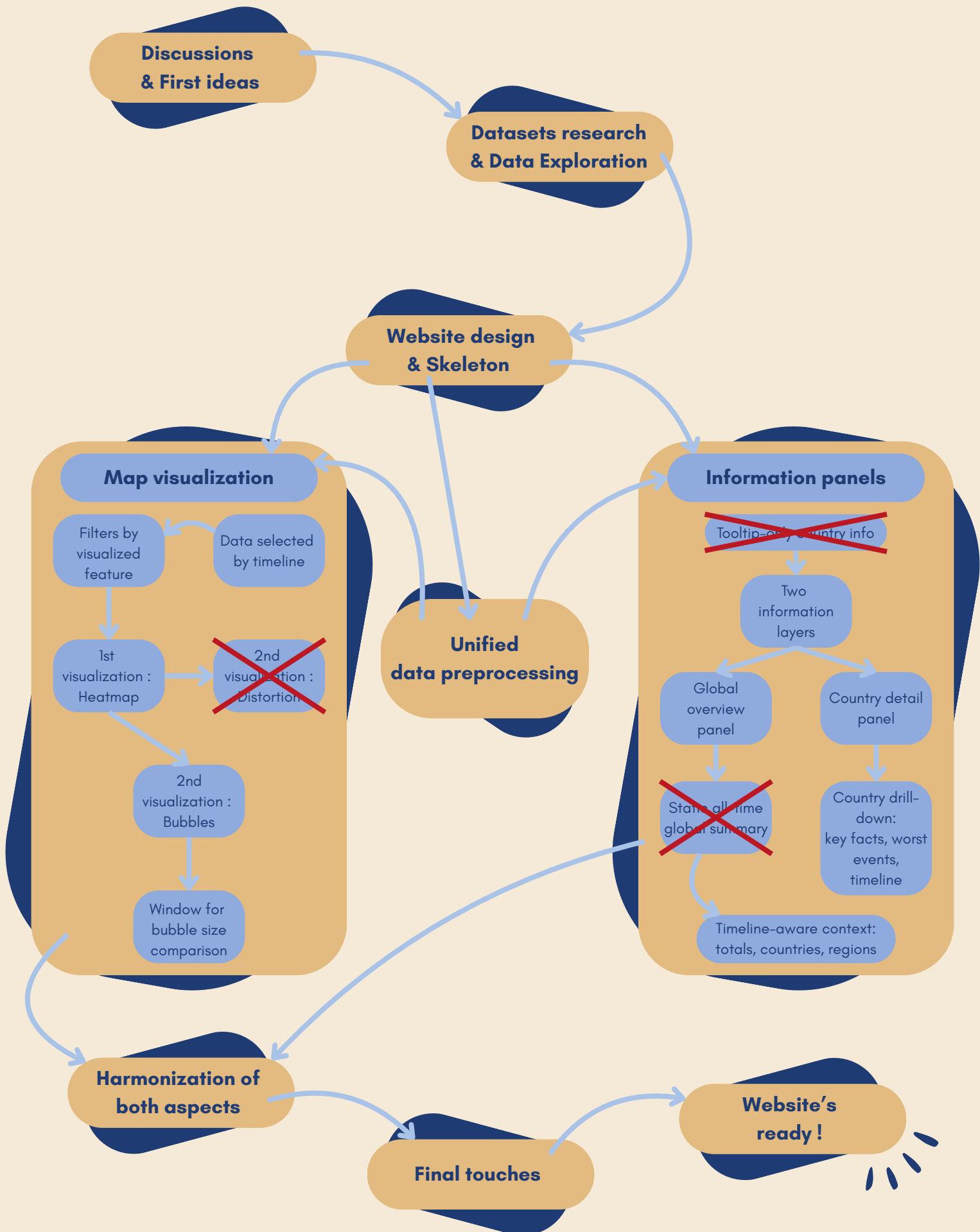


Process book

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I. Path : Evolution of our project



I. Path : Detailed explanation

Discussion & First ideas

The idea to work on natural disasters came from the will to treat them not as isolated cases as usually in the medias, but more as a systematic and scientific way. There was also a strong will to use map as visualization tool.

Dataset research & Data exploration

After some researches, we found the EMDAT dataset, that has natural disasters recorded since the beginning of the 20th century. Data exploration helped us understand better the dataset: features, missing values, ... This exploration mainly decided what we would visualize or not.

Website design & Skeleton

At this point, we had a clear idea of what we wanted to do and what was possible with our dataset. We decided to have a centre globe map with panels around, implementing two main features : visualization of global variables on the map, and country-specific information panel.

Map visualization

The idea of this big task was to monitor the visualization on the globe map. We first started by implementing the global structure and the different elements for this task. We created a timeline on bottom of the screen that allows to select a time range, and all the further functions take the time range as argument to take only data from this range. Then we created the side panel with clickable options corresponding to the features that we wanted to visualize. We first implemented the visualization by colorization of the map. Then we wanted to implement the visualization by distortion. We encountered some problems because we could not distort the map template. We could achieve some sort of distortion by superposing a distortable map over our globe map. We decided to change it to a bubble visualization because it was not easy to compare countries with this visualization. We added the bubble visualization with the same idea of superposition (except that we took a bubble shape instead of the country shape) and added a window to compare the countries of the same continent. This window opens when clicking on the continent background.

Information panels

The idea of this task was to design side panels for additional information that could not be clearly visualized on the map. We approached this task as an evolutive feature. We first considered a country specific tooltip that would open when hovering over a country. However, this idea failed quite quickly because the information we wanted to communicate was too rich for such a small interaction space. We then switched to side panels and separated the information into two levels. The first one was a global overview panel, which gives a general summary of the dataset with values such as the deadliest events, total number of events, most common disaster types, and affected regions. The second one was a country-specific panel, which appears after selecting a country and gives more detailed information such as the deadliest event in that country, the most common disaster types, and the evolution of disasters over time. The final step was to connect these panels with the timeline. As for the map visualization, the selected time range became the basis for computing the displayed information. This made the interface more coherent, because the map and the side panels now react to the same temporal selection.

Unified Data preprocessing

After choosing the general idea for the website design, we started dividing the work on the two independent features above. This is the moment where we felt the need for a unified data preprocessing. We chose to have a common JSON file with all data treatment done, from which both groups could start working to create the two features.

Harmonization of both aspects

As work was divided in two groups, some elements were off at the end : some behaviors were not expected in the same way by both groups, or seemed not intuitive for the other group. Some behavior even created conflict between the features, see "Challenges & Decisions" section for more details. This was solved by discussing each behavior and adding minor changes.

Final touches

At this point, the website was already functional, we just wanted some final touches on the style/appearance of the website :

- adapting the color palettes of the heatmaps
- adding animations on panel/window openings
- some small fixes, e.g. the timeline would update on every year tick and not on release

II. Challenges & Decisions

Distortion or bubbles ?

It was quite clear for us that the easiest way to visualize the features was colorization of the map. However, we wanted to visualize several features at a time. Our first idea was to distort the map according to the data : the bigger the feature the greater the size of the country. We encountered both a technical and visual issue : the map that we used could not be distorted easily, and when we succeeded to obtain a distorted overlay over the map, we realized that it was not easy to understand which country were bigger than others.

We decided to solve the problem by visualizing the feature via bubbles appearing over the countries. The idea stays the same : the bigger the feature for a country the greater the size of the bubble. We also added a window where bubbles of a same continent are clumped together to compare better the sizes.

What to click ?

After both main features (map visualization and per-country informations window) were finished, there was a confusion on what window should open in function of where the user clicked. When two features were visualized at the same time with bubbles, clicking on a continent would open the window to compare bubbles sizes (with bubbles clumped together). But at the same time, clicking on a country was supposed to open the country informations window.

This conflict was solved by deciding that clicking on a bubble would open the size comparison window, while clicking on a country on the background would open the country information panel.

Intuitiveness for the user

When finishing the map visualization and information panels, that were done in parallel by two groups, both groups had a review of the feature that they did not implement. The overall feedback was that the website was sometime unintuitive for some behaviors. Without explanation, it would not be very clear that the first feature was visualized by the colour and the second by the bubbles.

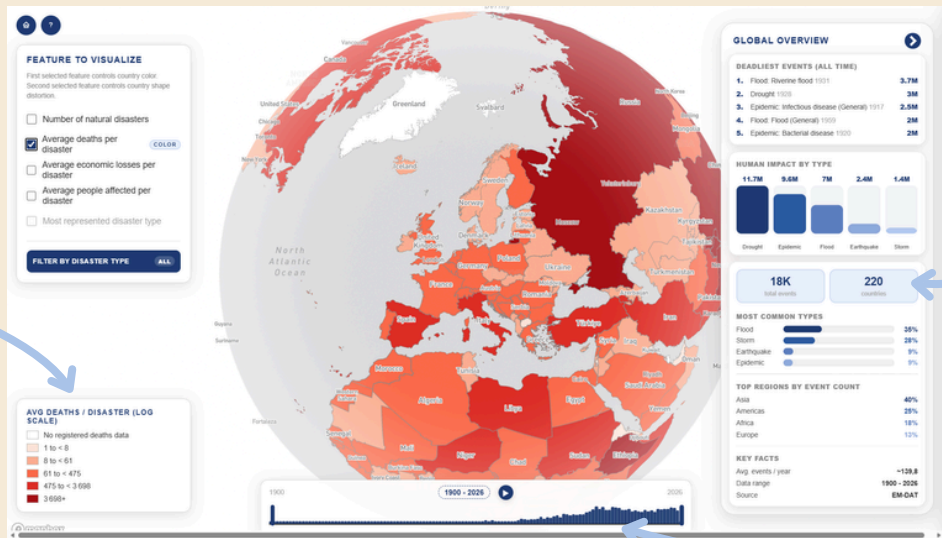
This issue was solved first by the "about" button, that provides explanations on how the website works. We also discussed the ways that we could make the website more intuitive, this is part of the step "harmonization of both aspects" in the previous section.

III. Sketches & Changes

Sketch from previous milestone, for home screen



Current home screen



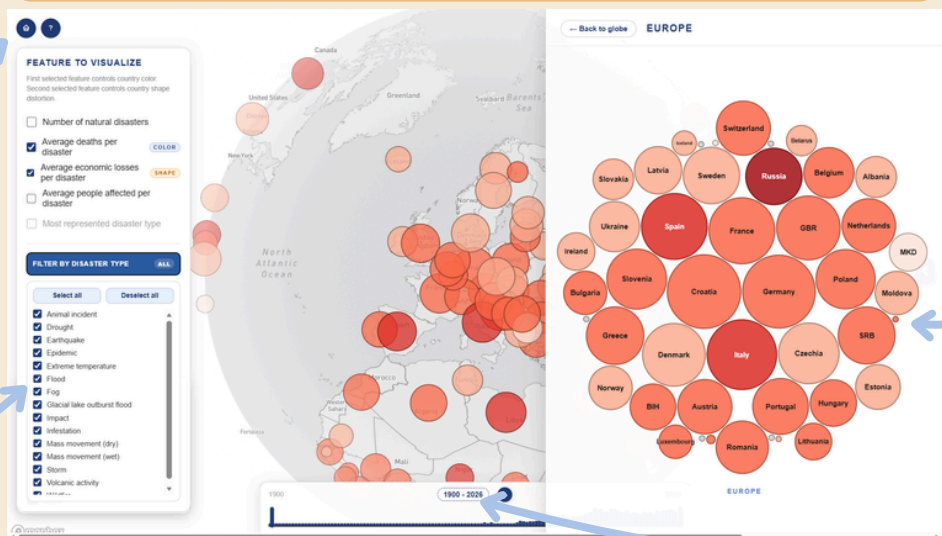
We added a more detailed legend, to make colours more meaningful. This was important also because of the use of log scale for a lot of features.

"Home" and "About" buttons:
The home button resets the visualization if user gets lost in what he's visualizing, and the about button opens a window with explanations

The global overview panel is in one scrollable block, no more "more informations" button with two different panels.

The timeline itself now conveys information: it contains the bar plot for the distribution of the number of natural events over time

New elements & other changes (screenshot from current website)



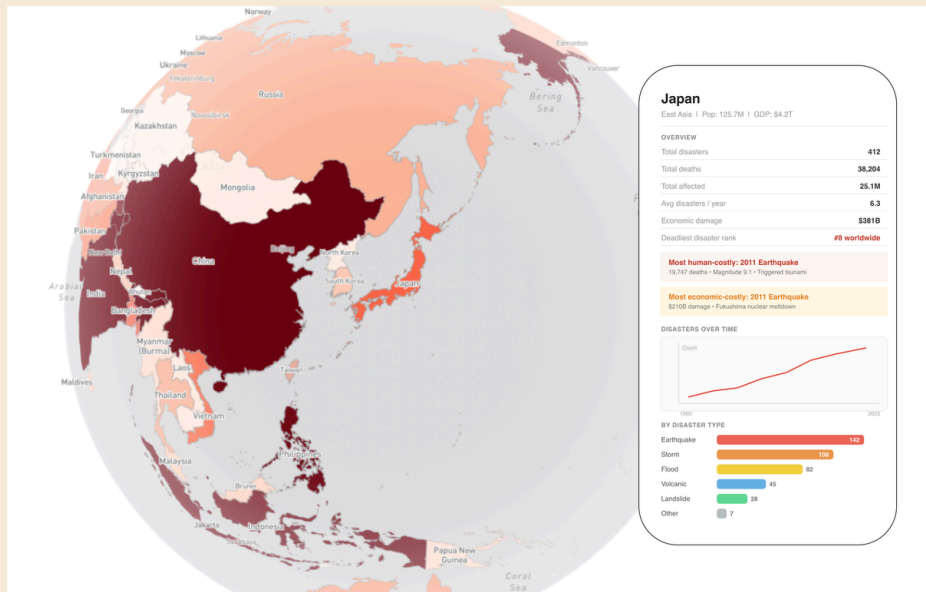
It was already planned to have a selection by disaster type, we changed it to be on a scrollable panel hidden by default.

We added a panel to compare more easily the sizes of the bubble than directly on the map.

There is possibility to select directly one unique year to visualize data from the same year, no need to select manually with timeline.

III. Sketches & Changes

Sketch from previous milestone, for country panel



Current home screen



The country specific panel is very close to what was planned : we made it scrollable and added an option to close it.

General comment on Plan vs Reality

In general, we achieved most of the ideas that we had for the website during the previous milestone. All the panels or features that we imagined are well implemented, we added some ideas such as distribution of natural disasters over time on the timeline.

The design of the elements is often different, due to our choice of style to harmonize the overall appearance of the website (different color palette, different font, ...)

IV. Peer assessment

Koorosh Sahafi

- Global overview panel
- Country information panel
- Panel interaction behavior
- Timeline driven information updates
- UI interaction refinement
- Contributed to data aggregation layer

Ella Neike

- Make the globe available using Mapbox
- First draft of an interactive map with clickable countries, a data panel, and a choropleth map visualization
- Prototype of a map distortion, later replaced by a bubble map
- Continent-level bubble panel with clickable bubbles
- Bubble legend with hover interaction
- Play animation on the timeline
- About and Home buttons
- UI interaction refinements
- Overall bug fixing

Timothée Coester

- Timeline structure and first design (later refined by Mohammadamin)
- Feature selection panel structure (without filters by disaster type, which was done by Ella)
- Visualization of features through map colorization
- Legend panel
- Process book
- Screencast

Mohammadamin Azarsa

- Data preprocessing pipeline
- Data aggregation layer
- Disaster-type filtering
- Timeline interaction refinements
- Interface animations and motion design
- Website deployment
- UI debugging

Common work

Work which is hard to attribute to a specific person, either because it was done in parallel, or has been done together in meetings :

- Exploratory data analysis
- Decisions about website design in general